Week 7 progressive hints

Use the hints in order. After each hint, return to your own work and continue until you reach a new, specific impasse. The last hint may give a nearly complete setup but does not give the final result.

W07-Q01-PA

1. With $h(x, y) = 12 - x - 2y$, use $\mathcal{L} = \log x + 2 \log y + \mu h(x, y)$.
2. Compare the two stationarity equations after accounting for the factor two in both the objective derivative with respect to y and the constraint derivative.

W07-Q01-PB

1. The logarithmic objective is strictly concave on the positive orthant, and the equality constraint is affine.
2. With resource a, first express both optimisers as functions of a. Substitute them into the objective, then compare $V'(a)$ with $\partial \mathcal{L}/\partial a$.

W07-Q02-PA

1. Write $\mathcal{L} = -(x-2)^2 - (y-1)^2 + \mu(a - x - y)$.
2. Express x and y in terms of μ, then impose $x + y = a$.
3. Substitute the optimisers into the two squared deviations. The objective's Hessian supplies a direct sufficiency argument.

W07-Q02-PB

1. Under $a - x - y = 0$, the partial derivative of the Lagrangian with respect to a is the multiplier.
2. Multiplying an equality constraint by -1 leaves its feasible set unchanged but changes the sign of the coefficient multiplying it in the Lagrangian.

W07-Q03-PA

1. Use one multiplier for $a - x - y - z = 0$ and another for $x - y = 0$. Check that their two gradient vectors are linearly independent.
2. Add or compare the first two stationarity equations, then use $x = y$. The third stationarity equation links z to the first multiplier.
3. Solve as a function of a, substitute into the objective, and compare the derivative of the resulting value function with the first multiplier.

W07-Q04-PA

1. Use $g_1(x) = x$ and $g_2(x) = 3 - x$, both written as non-negative constraints.
2. After stationarity, list the two primal inequalities, the two multiplier inequalities and one complementary-slackness equation for each constraint.

W07-Q04-PB

1. The unconstrained stationary point lies beyond the upper bound.
2. At a positive upper-bound solution, complementary slackness determines the multiplier on $x \geq 0$. Use stationarity for the remaining multiplier, then check its sign.
3. Strict concavity of the objective and convexity of the feasible interval turn a feasible Kuhn–Tucker point into the unique global solution.

W07-Q05-PA

1. Use $g_1(x, y) = x$, $g_2(x, y) = y$ and $g_3(x, y) = 4 - x - y$.
2. On the requested branch, complementary slackness sets the first two multipliers to zero while the third constraint holds with equality.
3. Equate the two expressions for the resource multiplier and combine the result with $x + y = 4$.

W07-Q05-PB

1. If the resource constraint is slack and both inputs are positive, the stationarity conditions reproduce the unconstrained candidate. Test its feasibility.
2. On $x = 0$, examine separately whether the resource is slack or binding; then do the symmetric case $y = 0$. A negative required multiplier rejects a branch.
3. The Hessian is a constant multiple of the identity matrix, and the feasible set is the intersection of three affine half-spaces.

W07-Q06-PA

1. The inequality $x \leq 1$ must be written $1 - x \geq 0$, and $x \geq 0$ is a separate constraint.
2. Under the course convention, both multiplier terms enter the Lagrangian with a plus sign.
3. The unconstrained maximiser is outside the interval. At the upper endpoint, complementary slackness sets the lower-bound multiplier to zero; stationarity determines the other multiplier.

W07-Q07-PA

1. Before observation, $\bar{X} + 1/n$ is random; after inserting the four data values, it is a number.
2. Use linearity of expectation and $\mathrm{Var}(\bar{X}) = \sigma^2/n$. Then use $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$.
3. To prove consistency, apply Chebyshev's inequality to $\bar{X} - \theta$ and account for the deterministic term $1/n$, or use the fact that the derived mean squared error tends to zero.

W07-Q08-PA

1. Equate the population first moment $\theta/2$ to the sample first moment $\bar{X}$.
2. For bias, take expectations of the resulting estimator. For consistency, combine the law of large numbers with the transformation used in the estimator.
3. The fitted uniform support is an interval beginning at zero. Compare its estimated upper endpoint with the largest observed value.

W07-Q09-PA

1. Independence gives $L(p; \mathbf{x}) = p^s(1 - p)^{n-s}$ on $p \in [0, 1]$.
2. For $0 < s < n$, differentiate $s \log p + (n - s) \log(1 - p)$ and check its curvature.
3. When $s = 0$ or $s = n$, the logarithmic interior calculation is not valid at the relevant endpoint. Inspect the likelihood directly on the closed parameter space.

W07-Q09-PB

1. Multiply the n exponential densities and state the observed-sample support condition before taking logs.
2. Let $T = \sum_i x_i$ and separate $T > 0$ from $T = 0$. For $T > 0$, differentiate $n \log \lambda - \lambda T$ and check both parameter-space boundaries; for $T = 0$, inspect the likelihood directly.
3. For method of moments, equate $1/\lambda$ to $\bar{X}$. Express both methods in terms of T and compare their finite-solution conditions.

W07-Q10-PA

1. The matrix is symmetric. For a 2×2 symmetric matrix, use the signs of the leading principal minors.
2. The Hessian of $q(\mathbf{x}) + \mathbf{b}^{\mathsf{T}}\mathbf{x} + c$ is twice the quadratic-form matrix.

W07-Q10-PB

1. Set the two components of the column gradient equal to zero.
2. The Hessian is constant. Its definiteness supplies a global conclusion on $\mathbb{R}^2$.

W07-Q10-PC

1. A mean of independent normal random variables is normal. Divide the individual variance by the sample size.
2. Standardise the two-sided event. Express the probability using $2\Phi(2) - 1$ before using the supplied CDF value.